Project Proposal: Sustainable Recipe Transformation Engine

Focus: Multi-Objective Optimization for Low-Carbon Culinary Intelligence

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1. Problem Statement

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We aim to build a system that takes an existing recipe and automatically transforms it into a more sustainable and healthier version through intelligent ingredient substitution.

The transformed recipe must successfully:

- Reduce the overall carbon footprint
- Maintain or improve nutritional value
- Preserve similar taste and flavor profiles
- Remain practical and accessible to cook

For example, the system may replace high-emission ingredients like **beef** or cream(with environmentally friendly alternatives like **mushrooms** that possess similar flavor characteristics and nutritional properties.

Beef (1kg)=19.54 Standard carbon footprint/kg

Mushroom (1kg)= 1.78 Standard carbon footprint/kg

Have **41 similar flavour molecules** one of the highest mentioned in the flavourDB

Addressing Existing Limitations: The existing Carbon Footprint project tracks emissions per recipe but offers no optimization. This project addresses that gap by building an active ingredient substitution engine that minimizes the carbon footprint while preserving flavor profile similarity using FlavorDB2 cosine similarity. This approach combines recipe data, flavor relationships, nutrition information, and optimization techniques to generate improved recipe alternatives automatically.

2. Methodology

Phase 1: Data Integration We will construct a comprehensive knowledge base by combining several high-quality datasets:

Recipe data → RecipeDB, **Recipe1M+**

Flavor relationship data → FlavorDB, **FooDB (foodb.ca)**

Nutrition data → USDA FoodData Central, OpenFoodFacts , **FooDB (foodb.ca)**

Ingredient substitution/co-occurrence data → RecipeDB and recipe datasets

Cooking instructions and metadata → **Recipe1M+**, RecipeDB

Sustainability-> Sustainable FoodDB and SU-EATABLE LIFE, for carbon and water footprint metrics

Ingredient similarity information → graph analysis from recipe datasets

Phase 2: Building the Ingredient Network To model how ingredients interact, we will construct a multi-layered graph: If you blindly swap ingredients based purely on carbon footprint, the engine might suggest cooking strawberries with garlic. To prevent this, you build a Co-occurrence Graph.

1. **Co-occurrence Graph:** Identifies which ingredients are commonly cooked together.
2. **Flavor Similarity Graph:** Maps ingredients that share similar chemical profiles.
3. **Substitution Graph:** Determines which ingredients can physically replace others (e.g., swapping butter for oil). This is where your FooDB/FlavorDB cosine similarity comes in. You want to replace Beef with something that shares its umami flavor molecules.

Phase 3: The Transformation Engine The core of the project is the multi-objective optimization engine. For instance, if a user inputs a beef stew recipe, the engine might suggest a mushroom and lentil blend as an optimal substitute.

$$Minimize\ e = w_1(\text{Carbon}) + w_2(\text{Flavor Divergence}) - w_3(\text{Nutrition})$$

(Note: We subtract Nutrition because in a "Minimization" equation, you subtract the things you want to maximize, and add the things you want to shrink!)

And update your constraints to remove the money aspect:

- Calories ≤ 2000
- Protein $\geq 60\text{g}$
- Sugar $\leq \text{limit}$
- $\text{Flavor Similarity Score} \geq 0.8$ *(Ensures the recipe still tastes good)*

Optimization Equation:

Type of optimization problem

Formulation	Type
linear objective + linear constraints	Linear Programming (LP)
integer recipe selections	Integer Linear Programming (ILP)
nonlinear nutrition/taste functions	Nonlinear Programming
convex objective + convex constraints	Convex Optimization
discrete substitutions + combinatorial search	Combinatorial Optimization

Phase 4: Explainability (XAI) The system will not simply output a list of new ingredients; it will explain its underlying logic to the user. For example: *"Lentils were chosen because they share umami compounds with beef but reduce carbon emissions by 82%."*

3. Technical Requirements

- **Languages:** Python (strong knowledge)
- **Machine Learning/Math:** PyTorch, Scikit-learn, DEAP (for optimization) **(just a Python library)**
- **Graph Processing:** NetworkX or Neo4j. (will do ml next sem)
- **Data Handling:** Pandas, MongoDB. (similar to microsoft excel - few week crash course)
- **Interface:** Streamlit or Flask (for a web-based dashboard).

Good python skills ,

Basic python libraries knowledge,
Good optimization knowledge,

Basic ML knowledge (will do ml next sem)**) ,**
Basic data handling.

4. Research Timeline

Semester	Focus	Key Deliverable
Semester 1	Data collection, cleaning, and graph construction.	A "Sustainability Scoring" tool for existing recipes.
Semester 2	Developing the substitution and optimization algorithms.	A working "Transformation Engine" prototype.
~~**Semester 3**~~	~~Building the dashboard, human testing, and final report.~~	~~Complete Research Thesis and Interactive App.~~ optional

Semester 1: The Setup & The Calculator

- **What you actually do:** You are just gathering the raw materials. You will download the datasets, clean up the messy data, and write the code to build your ingredient network graphs using your Discrete Structures knowledge.
- **The Key Deliverable:** A basic calculator. At the end of this semester, your code will only be able to look at a recipe and say, *"This recipe has a carbon footprint of X."* It **will not** suggest any substitutes or changes yet.

Semester 2: The Brains (The Transformation Engine)

- **What you actually do:** This is the heavy lifting. You will write the actual optimization math we just corrected to find the best ingredient swaps. This times out perfectly, because you will be learning the necessary ML concepts in class right as you need to code them.
- **The Key Deliverable:** The core engine. By the end of this semester, you will have a working Python script where you input a beef stew recipe, and the code successfully outputs a mushroom-lentil recipe with a lower carbon footprint.

Semester 3: The Presentation & The Thesis not required

- **What you actually do:** You make everything look good for the final grade. You will put a simple web interface (like Streamlit) on top of your Python script so people can click around and test it. You will also write up your entire research journey.

- **The Key Deliverable:** A finished, interactive web app and your final, formatted thesis document ready to be compiled in Overleaf and submitted to your professors.

5. Why This Research is Novel

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While many applications can score a recipe's carbon footprint, this project is unique because:

1. **Optimization Over Observation:** Unlike previous initiatives like the Carbon Footprint project that only track emissions, this engine actively recreates the recipe to minimize impact.
2. **Generative Change:** It doesn't just judge existing meals; it intelligently builds new ones.
3. **Flavor Preservation:** It utilizes quantitative molecular flavor profiles (via FlavorDB2 cosine similarity) to guarantee the revised version actually tastes good.
4. **Graph Intelligence:** It treats food as a complex system of networked relationships, aligning perfectly with the advanced computational gastronomy goals of CosyLab.

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6. Future Extensions

- **Budget Awareness:** Optimizing for the cheapest possible sustainable ingredients.
- **Regional Adaptation:** Suggesting substitutes based on local supply chain availability.
- **LLM Integration:** Utilizing Large Language Models to dynamically rewrite the cooking instructions to accommodate the newly substituted ingredients.